



*Building safer labs,
one hazard at a time.*

LabPilot AI

An intelligent safety co-pilot for every laboratory

TEAM

Fusion Minds

MEMBERS

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HACKATHON

AMD Developer Hackathon, 2026

Laboratory Accidents: A Global Safety Challenge

What problem exists?

A comprehensive analysis of 1,126 laboratory-associated incidents reported globally between 1900 and 2025 identified:

- **81 fatalities (7.2%)** • **148 outbreaks (13.3%)**

- Research laboratories: 276 infections, 8 fatalities

- Clinical laboratories: 227 infections, 5 deaths

Why it matters / Real-world impact

Traditional safety practice relies on static manuals and human memory — both prone to failure under pressure. The result: fatalities, injuries, disease outbreaks, costly equipment damage, research delays, and regulatory penalties. The WHO and ILO estimate approximately 2 million workers die annually from work-related causes.

30–60%

of clinical laboratory workers are affected by needlestick injuries.

67.5 / 1,000

exposure-accident rate per 1,000 active licenses (2024).

66.7%

of high-risk laboratory-acquired infections were caused by *Brucella melitensis*.

LabPilot AI: A Safety Co-Pilot For Every Lab

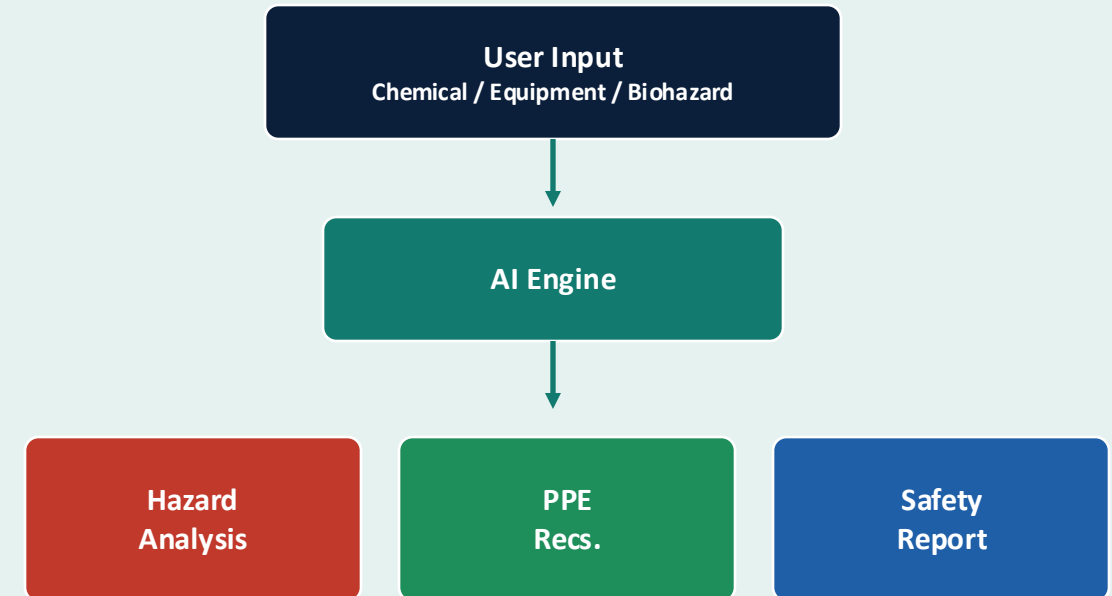
What is LabPilot AI?

LabPilot AI is an intelligent laboratory safety platform developed for the AMD Developer Hackathon. It helps students, researchers, and laboratory staff identify hazards, improve safety practices, and reduce laboratory accidents.

How does it solve the problem?

- Instant hazard identification: AI-powered chemical, biological, and equipment safety analysis.
- Intelligent recommendations: context-aware PPE and safety-protocol suggestions.
- Automated reporting: generates comprehensive safety reports on demand.
- 24/7 accessibility: web-based platform available anywhere, anytime.

Key idea



Identify hazards, assess risk, recommend PPE, and generate a compliance-ready report — all from one input.

What LabPilot AI Actually Does



Chemical Hazard Detection

Database of 100+ chemicals with hazard information



Biological Safety Guidance

Biosafety levels, risk groups & waste disposal



Physics & Equipment Safety

Safe operating procedures for lab equipment



AI Assistant

Safety score, AI guidance, intelligent recommendations



Risk Assessment

Evaluate risks and prioritize control measures



Safety Reports

Create and download detailed safety reports



Emergency Guidelines

Step-by-step emergency response procedures



Compliance Support

Ensure regulatory compliance and best practices

System Architecture

Components — what each stage does

- User — students, researchers, or lab staff access the platform from any device.
- Web Interface — a lightweight, responsive front-end for entering hazards and viewing results.
- Database — structured reference data on 100+ chemicals, PPE, and biosafety rules.
- AI Engine — an LLM that interprets input, checks it against the database, and generates guidance.
- Outputs — hazard analysis, PPE recommendations, and an auto-generated safety report.

User person accessing the platform

Web Interface (HTML / CSS / JS)

Database (Chemicals, PPE, Biosafety, Rules)

AI Engine (LLM Integration)

**Hazard
Analysis**

**Recommen-
dations**

**Report
Gen.**

Technology Stack

Frontend

HTML, CSS, JavaScript

AI Integration

Large Language Models (LLM) for safety analysis

Backend / Logic

GitHub for better development

Data Layer

Structured database — 100+ chemicals, PPE & biosafety rules

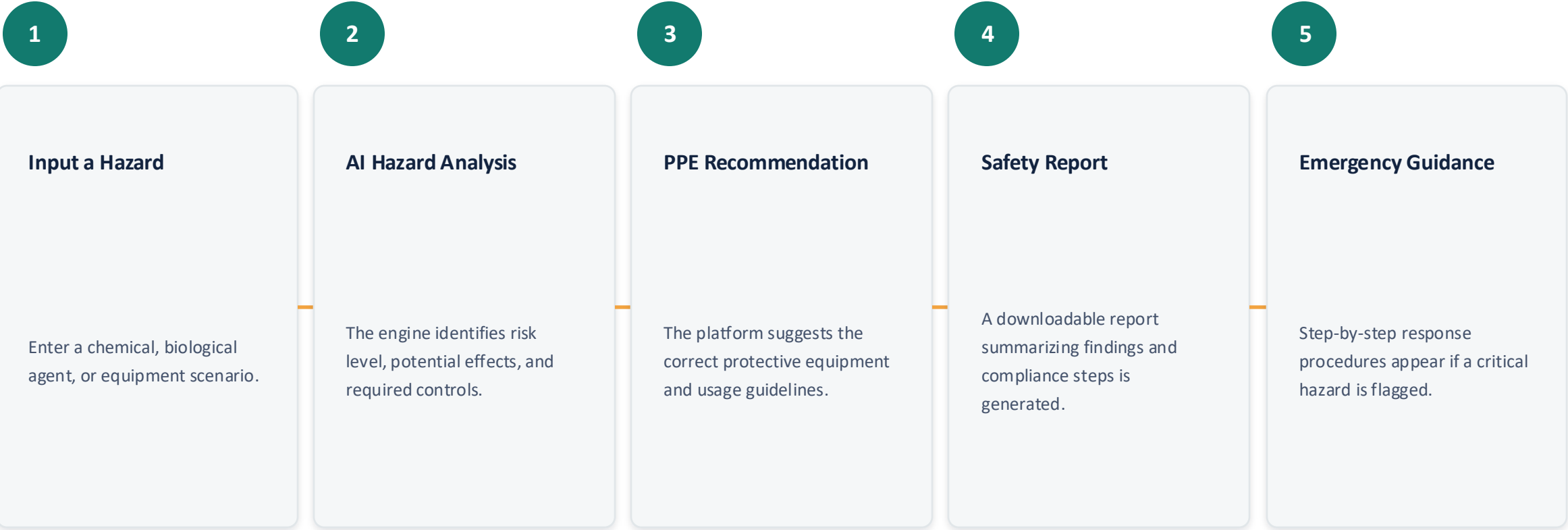
Hosting

GitHub

Key Technical Highlights

- Grounded AI recommendations in a curated database of 100+ chemicals — so outputs stay accurate instead of relying on the model's memory alone.
- Combined rule-based hazard classification with LLM-generated explanations for clarity.
- Designed for 24/7 web accessibility, so any lab member can check a hazard instantly without specialized software.

Live Demo Flow



What Makes LabPilot AI Different

The Old Way

- ✗ Static, printed safety manuals
- ✗ Reliance on memorized protocols
- ✗ Manual, delayed incident reporting
- ✗ One-size-fits-all PPE guidance
- ✗ No real-time risk scoring

The LabPilot AI Way

- ✓ Instant, AI-powered hazard lookup
- ✓ Context-aware PPE & protocol suggestions
- ✓ Automated, on-demand safety reports
- ✓ Personalized risk assessment per scenario
- ✓ 24/7 access from any device

Speed

Answers in seconds instead of manual manual look-ups.

Accuracy

Grounded in a structured 100+ chemical database.

Comprehensiveness

Covers chemical, biological, and equipment risk in one place.

Future Scope

1

Mobile App

Native iOS/Android version for on-the-go hazard checks.

2

Wearable Integration

Real-time alerts from smart PPE sensors.

3

Multi-language Support

Urdu and regional-language interfaces for wider adoption.

4

Incident Analytics

Trend tracking across labs and departments.

5

Regulatory Auto-Updates

Sync with evolving safety standards.

6

AR-Based Training

Augmented-reality walkthroughs for emergency drills.

Who This Helps

University Students

Safer, more confident hands-on lab experience during coursework.

Research Laboratories

Fewer infection and exposure incidents during experiments.

Clinical Laboratories

Lower needlestick and biohazard risk for technicians.

Lab Administrators

Easier compliance tracking and audit-ready reporting.

New / Junior Staff

Faster, guided onboarding to safety protocols.

Institutions

Reduced regulatory penalties and fewer research delays.

Team Contributions

Sawera Ansar

- Team leader • Fronted and UI developer • Research Analyst • Documentation Lead

Minahil Fatima

- Presentation lead • Data Collection lead

Simran sodaghar

- Video demo lead

Sadia Eman

Video demo lead

Challenges & Solutions

Structuring reliable hazard data

Built a curated dataset of 100+ chemicals instead of relying purely on model memory.

Keeping AI recommendations accurate

Grounded LLM outputs in the safety database rather than open-ended generation.

Making safety data approachable

Paired plain-language explanations with technical hazard classifications.

Coordinating a cross-disciplinary team

Divided work by domain — safety research, engineering, design — with regular syncs.

Bringing It Together

Summary

LabPilot AI is an AI-powered laboratory safety platform that identifies hazards, recommends the right PPE, and generates safety reports in real time — turning static manuals into an always-available safety co-pilot.

Expected Benefits

- Faster hazard identification and response
- Reduced lab accidents and exposure incidents
- Consistent regulatory compliance across shifts
- Less dependence on memorized protocols

Vision for the Future

We hope LabPilot AI becomes a standard companion in academic and research labs — helping build a culture where safety is proactive rather than reactive, one hazard at a time.



Thank You

Building safer labs, one hazard at a time.

Questions?

[Project link] <https://saweraansar.github.io/LabPilot-AI-2026/>